

Complex Number DPP

“Basic & Simplification Questions”

1. The value of i^{2014} is

- (a) i (b) $-i$
(c) 1 (d) -1

2. If $a < 0, b > 0$, then $\sqrt{a} \cdot \sqrt{b}$ is equal to

- (a) $-\sqrt{|a| \cdot b}$ (b) $\sqrt{|a| \cdot b} \cdot i$
(c) $\sqrt{|a|b}$ (d) None of these

3. The value of the sum $\sum_{n=1}^{13} (i^n + i^{n+1})$, where $i = \sqrt{-1}$ is

- (a) i (b) $i - 1$
(c) $-i$ (d) 0

4. The value of $\frac{2^n}{(1+i)^{2n}} + \frac{(1+i)^{2n}}{2^n}$, $n \in I$, is equal to

- (a) 0 (b) 2
(c) $\{1 + (-1)^n\} \cdot i^n$ (d) None of these

5. If $i = \sqrt{-1}$, then the number of values of $i^n + i^{-n}$ for different $n \in I$ is

- (a) 3 (b) 2
(c) 4 (d) 1

6. If $z = (-5i)\left(\frac{1}{8}i\right)$, then $\text{Im}(z)$ is equal to

- (a) 1 (b) 0
(c) -1 (d) None of these

7. If $4x + i(3x - y) = 3 + i(-6)$, where x and y are real numbers, then the value of x and y respectively are

- (a) 3.33 (b) $\frac{3}{5}, \frac{33}{5}$
(c) $\frac{3}{4}, \frac{33}{4}$ (d) None of these

8. The value of $3(7 + i7) + i(7 + i7)$ is

- (a) $15 + 27i$ (b) $14 + 28i$
(c) $14 - 28i$ (d) $14 + 23i$

9. The value of $\left[\left(\frac{1}{3} + i\frac{7}{3}\right) + \left(4 + i\frac{1}{3}\right)\right] - \left(-\frac{4}{3} + i\right)$ is

- (a) $\frac{5}{3} - i\frac{17}{3}$ (b) $\frac{17}{3} - i\frac{5}{3}$
(c) $\frac{17}{3} + i\frac{5}{3}$ (d) $\frac{17}{5} - i\frac{4}{3}$

10. The value of $\frac{z_1}{z_2}$, where $z_1 = 2 + 3i$ and $z_2 = 1 + 2i$, is

- (a) $\frac{8}{5} + \frac{1}{5}i$ (b) $\frac{8}{5} - \frac{1}{5}i$
(c) $\frac{1}{5} - \frac{8}{5}i$ (d) None of these

11. If $z + z^3 = 0$, then which of the following must be true on the complex plane?

- (a) $\text{Re}(z) < 0$ (b) $\text{Re}(z) = 0$
(c) $\text{Im}(z) = 0$ (d) $z^4 = 1$

12. The value of i^i is

- (a) $e^{-\pi/2}$ (b) $e^{\pi/2}$
(c) $e^{\pi/4}$ (d) None of these

13. The value of $\frac{i^{592} + i^{590} + i^{588} + i^{586} + i^{584}}{i^{582} + i^{580} + i^{578} + i^{576} + i^{574}} - 1$ is

- (a) -1 (b) -2
(c) -3 (d) -4

14. $i^{57} + \frac{1}{i^{125}}$, when simplified has the value

- (a) 0 (b) $2i$
(c) $-2i$ (d) 2

15. $i^n + i^{n+1} + i^{n+2} + i^{n+3}$ is equal to

- (a) 1 (b) -1
(c) 0 (d) None of these

16. $1 + i^2 + i^4 + i^6 + \dots + i^{2n}$ is

- (a) positive (b) negative
(c) 0 (d) Can't be determined

17. The value of $\left[i^{19} + \left(\frac{1}{i}\right)^{25}\right]^2$ is

- (a) 4 (b) -4
(c) 2 (d) -2

18. If n is any positive integer, then the value of $\frac{i^{4n+1} - i^{4n-1}}{2}$ equals

- (a) 1 (b) -1
(c) i (d) $-i$

19. For a positive integer n , the expression $(1 - i)^n \left(1 - \frac{1}{i}\right)^n$ equals

- (a) 0 (b) $2i^n$
(c) 2^n (d) 4^n

20. If the number $\frac{(1-i)^n}{(1+i)^{n-2}}$ is real and positive, then n is

- (a) any integer (b) 2λ
(c) $4\lambda + 1$ (d) None of these

21. The smallest positive integer n for which

$$\left(\frac{1+i}{1-i}\right)^n = -1 \text{ is}$$

- (a) 1 (b) 2
(c) 3 (d) 4



Complex Number DPP

22. The smallest positive number n for which

$$(1+i)^{2n} = (1-i)^{2n} \text{ is}$$

- (a) 4 (b) 8
(c) 2 (d) 12

23. The complex number $\frac{1+2i}{1-i}$ lies in the

- (a) I quadrant (b) II quadrant
(c) III quadrant (d) IV quadrant

24. The value of $\frac{(1-i)^3}{1-i^3}$ is equal to

- (a) i (b) -1
(c) 1 (d) -2

25. $\left(\frac{1+i}{1-i}\right)^2 + \left(\frac{1-i}{1+i}\right)^2$ is equal to

- (a) $2i$ (b) $-2i$
(c) -2 (d) 2

26. The value of $\operatorname{Re} \left[\frac{(1+i)^2}{3-i} \right]$ is equal to

- (a) $-\frac{1}{5}$ (b) $\frac{1}{5}$
(c) $\frac{1}{10}$ (d) $-\frac{1}{10}$

27. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$, then

- (a) $x = 3, y = 1$ (b) $x = 1, y = 3$
(c) $x = 0, y = 3$ (d) $x = 0, y = 0$

28. If z is a complex number such that $|z| \neq 0$ and $\operatorname{Re}(z) = 0$, then

- (a) $\operatorname{Re}(z^2) = 0$ (b) $\operatorname{Im}(z^2) = 0$
(c) $\operatorname{Re}(z^2) = \operatorname{Im}(z^2)$ (d) None of these

29. If $(x + iy)^{1/3} = a + ib$, then $\frac{x}{a} + \frac{y}{b}$ is equal to

- (a) $2(a^2 - b^2)$ (b) $4(a^2 - b^2)$
(c) $8(a^2 - b^2)$ (d) None of these

30. If $\left(\frac{1+i}{1-i}\right)^3 - \left(\frac{1-i}{1+i}\right)^3 = x + iy$, then (x, y) is equal to

- (a) $(0, 2)$ (b) $(-2, 0)$
(c) $(0, -2)$ (d) None of these

31. The value of $\left(\frac{1+i}{\sqrt{2}}\right)^8 + \left(\frac{1-i}{\sqrt{2}}\right)^8$ is equal to

- (a) 4 (b) 6
(c) 8 (d) 2

32. If $z = x - iy$ and $z^{1/3} = p + iq$, then

$$\left(\frac{x}{p} + \frac{y}{q}\right) / (p^2 + q^2) \text{ is equal to}$$

- (a) 1 (b) -1
(c) 2 (d) -2

33. If $\left(\frac{1+i}{1-i}\right)^x = 1$, then

- (a) $x = 4n$, where n is any positive integer
(b) $x = 2n$, where n is any positive integer
(c) $x = 4n + 1$, where n is any positive integer
(d) $x = 2n + 1$, where n is any positive integer

34. If $z_1 = 2\sqrt{2}(1+i)$ and $z_2 = 1 + i\sqrt{3}$, then $z_1^2 z_2^3$ is equal to

- (a) $128i$ (b) $64i$
(c) $-64i$ (d) None of these

35. The value of $(1+i)^3 + (1-i)^3$ is equal to

- (a) 1 (b) -2
(c) 0 (d) -4

36. The sequence $S = i + 2i^2 + 3i^3 + 4i^4 + \dots$ upto 100 terms simplifies to, where $i = \sqrt{-1}$

- (a) $50(1-i)$ (b) $25i$
(c) $25(1+i)$ (d) $100(1-i)$

37. A value of θ for which $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ is purely imaginary, is

- (a) $\frac{\pi}{3}$ (b) $\frac{\pi}{6}$
(c) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (d) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

38. The real values of x and y for which the following equality hold, are respectively

$$(x^4 + 2xi) - (3x^2 + iy) = (3 - 5i) + (1 + 2iy)$$

- (a) 2, 3 or $-2, 1/3$ (b) 1, 3 or $-1, 1/3$
(c) $2, 1/3$ or $-2, 3$ (d) None of these

39. Let $\left(-2 - \frac{1}{3}i\right)^3 = \frac{x+iy}{27}$ ($i = \sqrt{-1}$), where x and y are real numbers, then $y - x$ equals

- (a) 91 (b) 85
(c) -85 (d) -91

40. If $a = \cos\theta + i\sin\theta$, then $\frac{1+a}{1-a}$ is equal to

- (a) $i\cot\frac{\theta}{2}$ (b) $i\tan\frac{\theta}{2}$
(c) $i\cos\frac{\theta}{2}$ (d) $i\operatorname{cosec}\frac{\theta}{2}$

41. Real part of $\frac{1}{1-\cos\theta+i\sin\theta}$ is

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) $\frac{1}{2}\tan\theta/2$ (d) 2

42. The value of $x^4 + 9x^3 + 35x^2 - x + 4$ for $x = -5 + 2\sqrt{-4}$ is



Complex Number DPP

- (a) 0 (b) -160
(c) 160 (d) -164

“Conjugate of Complex Number”

43. If $z_1 = 9y^2 - 4 - 10ix$ and $z_2 = 8y^2 + 20i$, where $z_1 = \bar{z}_2$, then $z = x + iy$ is equal to

- (a) $-2 + 2i$ (b) $-2 \pm 2i$
(c) $-2 \pm i$ (d) None of the above

44. If $(1 + i)z = (1 - i)\bar{z}$, then z is

- (a) $x(1 - i), x \in R$ (b) $x(1 + i), x \in R$
(c) $\frac{x}{1+i}, x \in R^+$ (d) None of the above

45. If $z = \left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^5 + \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^5$, then

- (a) $\operatorname{Re}(z) = 0$
(b) $\operatorname{Im}(z) = 0$
(c) $\operatorname{Re}(z) > 0, \operatorname{Im}(z) > 0$
(d) $\operatorname{Re}(z) > 0, \operatorname{Im}(z) < 0$

46. Statement I : $3 + ix^2y$ and $x^2 + y + 4i$ are complex conjugate numbers, then $x^2 + y^2 = 4$.

Statement II : If sum and product of two complex numbers is real, then they are conjugate complex number.

Which of the above statement(s) is/are correct ?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

47. If $a^2 + b^2 = 1$, then $\frac{(1+b+ia)}{(1+b-ia)}$ is equal to

- (a) 1 (b) 2
(c) $b + ia$ (d) $a + ib$

48. If $z = x + iy$ lies in III quadrant, then $\frac{\bar{z}}{z}$ also lies in III quadrant, if

- (a) $x > y > 0$ (b) $x < y < 0$
(c) $y < x < 0$ (d) $y > x > 0$

49. The conjugate complex number of $\frac{2-i}{(1-2i)^2}$ is

- (a) $\frac{2}{25} + \frac{11}{25}i$ (b) $\frac{2}{25} - \frac{11}{25}i$
(c) $-\frac{2}{25} + \frac{11}{25}i$ (d) $-\frac{2}{25} - \frac{11}{25}i$

50. The conjugate of a complex number is $\frac{1}{i-1}$. Then, that complex number is

- (a) $-\frac{1}{i-1}$ (b) $\frac{1}{i+1}$
(c) $-\frac{1}{i+1}$ (d) $\frac{1}{i-1}$

51. If the conjugate of $(x + iy)(1 - 2i)$ is $1 + i$, then

- (a) $x - iy = \frac{1-i}{1-2i}$ (b) $x + iy = \frac{1-i}{1-2i}$
(c) $x = \frac{1}{5}$ (d) $x = -\frac{1}{5}$

52. If $z = \frac{4}{1-i}$, then $\bar{z}$ is (where, $\bar{z}$ is complex conjugate of z)

- (a) $2(1 + i)$ (b) $(1 + i)$
(c) $\frac{2}{1-i}$ (d) $\frac{4}{1-i}$

53. The complex numbers $\sin x + i \cos 2x$ and $\cos x - i \sin 2x$ are conjugate to each other for

- (a) $x = n\pi$ (b) $x = \left(n + \frac{1}{2}\right)\pi$
(c) $x = 0$ (d) No value of x

“Modulus of Complex Number”

54. If z is a complex number satisfying the relation $|z + 1| = z + 2(1 + i)$, then z is

- (a) $\frac{1}{2}(1 + 4i)$ (b) $\frac{1}{2}(3 + 4i)$
(c) $\frac{1}{2}(1 - 4i)$ (d) $\frac{1}{2}(3 - 4i)$

55. If $z = 1 + i \tan \alpha$, where $\pi < \alpha < \frac{3\pi}{2}$, then $|z|$ is equal to

- (a) $\sec \alpha$ (b) $-\sec \alpha$
(c) $\operatorname{cosec} \alpha$ (d) None of the above

56. The complex number z satisfies $z + |z| = 2 + 8i$.

Then, the value of $|z|$ is

- (a) 10 (b) 13
(c) 17 (d) 23

57. Consider two complex numbers α and β as

$$\alpha = \left(\frac{a+bi}{a-bi}\right)^2 + \left(\frac{a-bi}{a+bi}\right)^2, \text{ where } a, b \in R \text{ and}$$

$$\beta = \frac{z-1}{z+1}, \text{ where } |z| = 1, \text{ then}$$

- (a) both α and β are purely real
(b) both α and β are purely imaginary
(c) α is purely real and β is purely imaginary
(d) β is purely real and α is purely imaginary



Complex Number DPP

58. If $x - iy = \sqrt{\frac{a-ib}{c-id}}$, then $(x^2 + y^2)^2$ is equal to

- (a) $\frac{a^2-b^2}{c^2-d^2}$ (b) $\frac{a^2+b^2}{c^2+d^2}$
(c) $\frac{a^2+b^2}{c^2-d^2}$ (d) None of these

59. If α, β are two complex numbers, then $|\alpha|^2 + |\beta|^2$ is equal to

- (a) $\frac{1}{2}(|\alpha + \beta|^2 - |\alpha - \beta|^2)$
(b) $\frac{1}{2}(|\alpha + \beta|^2 + |\alpha - \beta|^2)$
(c) $|\alpha + \beta|^2 + |\alpha - \beta|^2$
(d) None of these

60. Let z be a complex number satisfying the equation $(z^3 + 3)^2 = -16$, then $|z|$ has the value equal to

- (a) $5^{1/2}$ (b) $5^{1/3}$
(c) $5^{2/3}$ (d) 5

61. If z is a complex number satisfying $z^4 + z^3 + 2z^2 + z + 1 = 0$, then $|z|$ is equal to

- (a) $\frac{1}{2}$ (b) $\frac{3}{4}$
(c) 1 (d) None of these

62. If $\frac{1-ia}{1+ia} = A + iB$, then $A^2 + B^2$ equals

- (a) 1 (b) α^2
(c) -1 (d) $-\alpha^2$

63. If z_1 and z_2 are two complex numbers satisfying the equation $\left|\frac{z_1+iz_2}{z_1-iz_2}\right| = 1$, then $\frac{z_1}{z_2}$ is

- (a) purely real (b) of unit modulus
(c) purely imaginary (d) None of the above

64. If $8iz^3 + 12z^2 - 18z + 27i = 0$, then

- (a) $|z| = \frac{3}{2}$ (b) $|z| = \frac{2}{3}$
(c) $|z| = 1$ (d) $|z| = \frac{3}{4}$

65. The value of $\left|\frac{1+i\sqrt{3}}{(1+\frac{1}{i+1})^2}\right|$ is

- (a) 20 (b) 9
(c) $\frac{5}{4}$ (d) $\frac{4}{5}$

66. If $z_1 = \sqrt{2}\left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4}\right)$ and $z_2 = \sqrt{3}\left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}\right)$, then $|z_1 z_2|$ is

- (a) 6 (b) $\sqrt{6}$
(c) $\sqrt{2}$ (d) $\sqrt{3}$

67. $\left|(1+i)\left(\frac{2+i}{3+i}\right)\right|$ is equal to

- (a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) -1

68. If $\frac{1-ix}{1+ix} = a - ib$ and $a^2 + b^2 = 1$, where a and b are real, then x is equal to

- (a) $\frac{2a}{(1+a)^2+b^2}$ (b) $\frac{2b}{(1+a)^2+b^2}$
(c) $\frac{2a}{(1+b)^2+a^2}$ (d) $\frac{2b}{(1+b)^2+a^2}$

69. The modulus of $\frac{(3+2i)^2}{(4-3i)}$ is

- (a) $\frac{13}{5}$ (b) $\frac{11}{5}$
(c) $\frac{9}{5}$ (d) $\frac{7}{5}$

70. If $x + iy = \frac{u+iv}{u-iv}$, then $x^2 + y^2$ is equal to

- (a) 1 (b) -1
(c) 0 (d) None of these

71. If z is a complex number, then

- (a) $|z^2| > |z|^2$ (b) $|z^2| = |z|^2$
(c) $|z^2| < |z|^2$ (d) $|z^2| \geq |z|^2$

72. If $z_1 \neq z_2$ and $|z_1 + z_2| = \left|\frac{1}{z_1} + \frac{1}{z_2}\right|$, then

- (a) atleast one of z_1, z_2 is unimodular
(b) $z_1 \cdot z_2$ is unimodular
(c) $z_1 \cdot z_2$ is non-unimodular
(d) None of the above

73. If z satisfies the equation $|z| - z = 1 + 2i$, then z is equal to

- (a) $\frac{3}{2} + 2i$ (b) $\frac{3}{2} - 2i$
(c) $2 - \left(\frac{3}{2}\right)i$ (d) $2 + \left(\frac{3}{2}\right)i$

74. If $z = \frac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2}$, then $|z|$ is equal to

- (a) 8 (b) 2
(c) 5 (d) None of these

75. If z_1, z_2 and z_3 are complex numbers such that

$$|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right| = 1, \text{ then}$$

$|z_1 + z_2 + z_3|$ is

- (a) 3 (b) 1
(c) greater than 3 (d) less than 1

76. If $(\sqrt{3}i + 1)^{100} = 2^{99}(a + ib)$, then $a^2 + b^2$ is equal to

- (a) 4 (b) 3
(c) 2 (d) 0



Complex Number DPP

77. If $w = \alpha + i\beta$, where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\left(\frac{w-\bar{w}z}{1-z}\right)$ is purely real, then the set of values of z is

- (a) $|z| = 1, z \neq 1$ (b) $|z| = 1$ and $z \neq 1$
(c) $z = \bar{z}$ (d) None of these

78. If $\frac{z-\alpha}{z+\alpha}$ ($\alpha \in R$) is a purely imaginary number and $|z| = 2$, then a value of α is

- (a) $\sqrt{2}$ (b) $\frac{1}{2}$
(c) 1 (d) 2

79. If $\text{Re}(z^2) = 0, |z| = 2$, then

- (a) $z = \pm\sqrt{2} \pm i\sqrt{2}$ (b) $z = \pm\sqrt{3} \pm i\sqrt{3}$
(c) $z = \pm\sqrt{3} \pm i\sqrt{2}$ (d) None of these

80. If $\frac{z-1}{z+1}$ is a purely imaginary number ($z \neq -1$), then the value of $|z|$ is

- (a) -1 (b) 1
(c) 2 (d) -2

81. If $z + \sqrt{2}|z + 1| + i = 0$, then z equals

- (a) $2 + i$ (b) $-2 + i$
(c) $-\frac{1}{2} + i$ (d) $-2 - i$

82. If $f(z) = \frac{7-z}{1-z^2}$, where $z = 1 + 2i$, then $|f(z)|$ is

- (a) $\frac{|z|}{2}$ (b) $|z|$
(c) $2|z|$ (d) None of these

"Multiplicative Inverse"

83. The multiplicative inverse of $4 - 3i$ is

- (a) $\frac{4}{25} - \frac{3i}{25}$ (b) $\frac{4}{25} + \frac{3i}{25}$
(c) $\frac{4}{16} + \frac{3i}{25}$ (d) None of these

84. If $z = 1 + i$, then the multiplicative inverse of z^2 is

- (a) $1 - i$ (b) $\frac{i}{2}$
(c) $-\frac{i}{2}$ (d) $2i$

85. The multiplicative inverse of $(6 + 5i)^2$ is

- (a) $\frac{11}{60} - \frac{60}{61}i$ (b) $\frac{11}{61} - \frac{60}{61}i$
(c) $\frac{9}{61} - \frac{60}{61}i$ (d) None of these

86. The multiplicative inverse of $\frac{3+4i}{4-5i}$ is

- (a) $-\frac{8}{25} + \frac{31}{25}i$ (b) $\frac{8}{25} - \frac{31}{25}i$
(c) $-\frac{8}{25} - \frac{31}{25}i$ (d) None of these

"Square Root of Complex Number"

87. The value of $\sqrt{-8 - 6i}$ is equal to

- (a) $1 + 3i$ (b) $\pm(1 - 3i)$
(c) $\pm(1 + 3i)$ (d) $\pm(3 - i)$

88. The value of $(4 + 3\sqrt{-20})^{1/2} + (4 - 3\sqrt{-20})^{1/2}$ is

- (a) ± 6 (b) 0
(c) $\pm\sqrt{5}$ (d) ± 3

89. Consider the following statements

If $x, y \in R$, then

I. $\sqrt{x + iy} + \sqrt{x - iy} = \sqrt{2(\sqrt{x^2 + y^2} + x)}$

II. $\sqrt{x + iy} - \sqrt{x - iy} = i\sqrt{2(\sqrt{x^2 + y^2} - x)}$

Which of the above statement(s) is/are correct ?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

90. $\sqrt{2}i$ equals

- (a) $1 + i$ (b) $1 - i$
(c) $-\sqrt{2}i$ (d) None of these

91. The value of $|\sqrt{2}i - \sqrt{-2}i|$ is

- (a) 2 (b) $\sqrt{2}$
(c) 0 (d) $2\sqrt{2}$

92. The square roots of $-2 + 2\sqrt{3}i$ are

- (a) $\pm(1 + \sqrt{3}i)$ (b) $\pm(1 - \sqrt{3}i)$
(c) $\pm(-1 + \sqrt{3}i)$ (d) None of these

93. The imaginary part of

$(3 + 2\sqrt{-54})^{1/2} - (3 - 2\sqrt{-54})^{1/2}$ can be

- (a) $-\sqrt{6}$ (b) $-2\sqrt{6}$
(c) 6 (d) $\sqrt{6}$

94. The square root of $7 + 24i$ are

- (a) $\pm(3 + 4i)$ (b) $\pm(3 - 4i)$
(c) $\pm(4 + 3i)$ (d) $\pm(4 - 3i)$

"Argument of Complex Number"

95. The modulus and argument of the complex number $\frac{1+2i}{1-3i}$ is



Complex Number DPP

- (a) $\frac{1}{\sqrt{2}}, \frac{3\pi}{4}$ (b) $\frac{1}{\sqrt{2}}, -\frac{3\pi}{4}$
 (c) $\frac{1}{2}, \frac{3\pi}{4}$ (d) None of the above

96. Let z and w be two non-zero complex numbers, such that $|z| = |w|$ and $\arg(z) + \arg(w) = \pi$. Then, z is equal to

- (a) w (b) $-w$
 (c) $-\bar{w}$ (d) $\bar{w}$

97. If $z = \frac{\sqrt{3}+i}{\sqrt{3}-i}$, then the fundamental amplitude of z is

- (a) $-\frac{\pi}{3}$ (b) $\frac{\pi}{3}$
 (c) $\frac{\pi}{6}$ (d) None of these

98. If z_1, z_2 and z_3, z_4 are two pairs of conjugate complex numbers, then

$\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ equals to

- (a) 0 (b) $\frac{\pi}{2}$
 (c) $\frac{3\pi}{2}$ (d) π

99. If $\frac{1+2i}{2+i} = r(\cos \theta + i \sin \theta)$, then

- (a) $r = 1, \theta = \tan^{-1} \frac{3}{4}$
 (b) $r = \sqrt{5}, \theta = \tan^{-1} \frac{4}{3}$
 (c) $r = 1, \theta = \tan^{-1} \frac{4}{3}$
 (d) None of these

100. If z_1 and z_2 are two non-zero complex numbers, such that $|z_1 + z_2| = |z_1| + |z_2|$, then $\arg(z_1) - \arg(z_2)$ is equal to

- (a) $-\pi$ (b) $-\frac{\pi}{2}$
 (c) 0 (d) $\frac{\pi}{2}$

101. If $z = 1 - \sin \alpha + i \cos \alpha$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$, then the modulus and the principal value of the argument of z are respectively

- (a) $\sqrt{2(1 - \sin \alpha)}, \left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$
 (b) $\sqrt{2(1 - \sin \alpha)}, \left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$
 (c) $\sqrt{2(1 + \sin \alpha)}, \left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$
 (d) $\sqrt{2(1 + \sin \alpha)}, \left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$

102. The principal argument of the complex number $\frac{(1+i)^5(1+\sqrt{3}i)^2}{2i(-\sqrt{3}+i)}$ is

- (a) $\frac{19\pi}{12}$ (b) $-\frac{7\pi}{12}$
 (c) $-\frac{5\pi}{12}$ (d) $\frac{5\pi}{12}$

103. The complex number $i + \sqrt{3}$ in polar form can be written as

- (a) $\frac{1}{\sqrt{2}}\left(\sin \frac{\pi}{6} + i \cos \frac{\pi}{6}\right)$ (b) $2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$
 (c) $\frac{1}{2}\left(\sin \frac{\pi}{6} + i \cos \frac{\pi}{6}\right)$ (d) $4\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$

104. If $\sqrt{3} + i = (a + ib)(c + id)$, then $\tan^{-1} \frac{b}{a} + \tan^{-1} \frac{d}{c}$ is equal to

- (a) $\frac{\pi}{2}$ (b) $-\frac{\pi}{3}$
 (c) $\frac{\pi}{6}$ (d) $-\frac{\pi}{6}$

105. If z is a complex number of unit modulus and argument θ , then $\arg\left(\frac{1+z}{1+\bar{z}}\right)$ equals

- (a) $-\theta$ (b) $\frac{\pi}{2} - \theta$
 (c) θ (d) $\pi - \theta$

106. If $\arg(z - 1) = \arg(z + 3i)$, then $x - 1 : y$ is equal to

- (a) 3:1 (b) 1:3
 (c) 3:2 (d) 2:3

“De-Moivre's Theorem”

107. If $x_r = \cos\left(\frac{\pi}{2^r}\right) + i \sin\left(\frac{\pi}{2^r}\right)$, then the value of $x_1 x_2 x_3 \dots \infty$ is

- (a) -1 (b) 1
 (c) 0 (d) None of these

108. $\frac{(\cos \theta + i \sin \theta)^4}{(\sin \theta + i \cos \theta)^5}$ is equal to

- (a) $\cos \theta - i \sin \theta$ (b) $\cos 9\theta - i \sin 9\theta$
 (c) $\sin \theta - i \cos \theta$ (d) $\sin 9\theta - i \cos 9\theta$

109. The expression $\left[\frac{1 + \sin \frac{\pi}{8} + i \cos \frac{\pi}{8}}{1 + \sin \frac{\pi}{8} - i \cos \frac{\pi}{8}}\right]^8$ is equal to

- (a) 1 (b) -1
 (c) i (d) $-i$

110. If $z = \frac{\cos \theta + i \sin \theta}{\cos \theta - i \sin \theta}$, $\frac{\pi}{4} < \theta < \frac{\pi}{2}$.

Then, $\arg(z)$ is

- (a) 2θ (b) $2\theta - \pi$
 (c) $\pi + 2\theta$ (d) None of these



Complex Number DPP

111. If $a = \cos \alpha + i \sin \alpha$, $b = \cos \beta + i \sin \beta$, $c = \cos \gamma + i \sin \gamma$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 1$, then $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha)$ is equal to

- (a) $\frac{3}{2}$ (b) $-\frac{3}{2}$
(c) 0 (d) 1

112. If $2\cos \theta = x + \frac{1}{x}$ and $2\cos \phi = y + \frac{1}{y}$, then

- (a) $\frac{x}{y} + \frac{y}{x} = 2\cos(\theta + \phi)$
(b) $x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\theta + n\phi)$
(c) $\frac{x^m}{y^n} + \frac{y^n}{x^m} = 2\cos(m\theta + n\phi)$
(d) $xy + \frac{1}{xy} = 2\cos(\theta - \phi)$

113. The value of $\frac{(\sin \frac{\pi}{8} + i \cos \frac{\pi}{8})^8}{(\sin \frac{\pi}{8} - i \cos \frac{\pi}{8})^8}$ is

- (a) -1 (b) 0
(c) 1 (d) $2i$

114. If $a = \cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3}$, then the value of $\left(\frac{1+a}{2}\right)^{3n}$ is

- (a) $(-1)^n$ (b) $\frac{(-1)^n}{2^{3n}}$
(c) $\frac{1}{2^{3n}}$ (d) $(-1)^n + 1$

115. The value of $\frac{4(\cos 75^\circ + i \sin 75^\circ)}{0.4(\cos 30^\circ + i \sin 30^\circ)}$ is

- (a) $\frac{10}{\sqrt{2}}(1+i)$ (b) $\frac{10}{\sqrt{2}}(1-i)$
(c) $\frac{5}{\sqrt{2}}(1+i)$ (d) None of these

116. The value of $(\cos 2\theta - i \sin 2\theta)^7 (\cos 3\theta + i \sin 3\theta)^{-5}$ is $(\cos 4\theta + i \sin 4\theta)^{12} (\cos 5\theta + i \sin 5\theta)^{-6}$

- (a) $\cos 33\theta + i \sin 33\theta$
(b) $\cos 33\theta - i \sin 33\theta$
(c) $\cos 47\theta + i \sin 47\theta$
(d) $\cos 47\theta - i \sin 47\theta$

117. $(\cos 2\theta + i \sin 2\theta)^{-5} (\cos 3\theta - i \sin 3\theta)^6 (\sin \theta - i \cos \theta)^3$ is equal to

- (a) $\cos 25\theta + i \sin 25\theta$
(b) $\cos 25\theta - i \sin 25\theta$
(c) $\sin 25\theta + i \cos 25\theta$
(d) $\sin 25\theta - i \cos 25\theta$

118. If $a = \cos 2\alpha + i \sin 2\alpha$, $b = \cos 2\beta + i \sin 2\beta$, $c = \cos 2\gamma + i \sin 2\gamma$ and $d = \cos 2\delta +$

$i \sin 2\delta$, then $\sqrt{abcd} + \frac{1}{\sqrt{abcd}}$ is equal to

- (a) $\sqrt{2} \cos(\alpha + \beta + \gamma + \delta)$
(b) $2 \cos(\alpha + \beta + \gamma + \delta)$
(c) $\cos(\alpha + \beta + \gamma + \delta)$
(d) None of these

119. The value of $\left(\frac{1 + \sin \frac{2\pi}{9} + i \cos \frac{2\pi}{9}}{1 + \sin \frac{2\pi}{9} - i \cos \frac{2\pi}{9}}\right)^3$ is

- (a) $-\frac{1}{2}(\sqrt{3} - i)$ (b) $-\frac{1}{2}(1 - i\sqrt{3})$
(c) $\frac{1}{2}(\sqrt{3} - i)$ (d) $\frac{1}{2}(1 - i\sqrt{3})$

"Cube Roots of Unity"

120. The value of $\text{amp}(i\omega) + \text{amp}(i\omega^2)$, where $i = \sqrt{-1}$ and $\omega = \sqrt[3]{1} = \text{non-real}$, is

- (a) 0 (b) $\frac{\pi}{2}$
(c) π (d) None of these

121. If $x^2 - x + 1 = 0$, then the value of

$\sum_{n=1}^5 \left(x^n + \frac{1}{x^n}\right)^2$ is

- (a) 8 (b) 10
(c) 12 (d) None of the above

122. If $3^{49}(x + iy) = \left(\frac{3}{2} + \frac{\sqrt{3}}{2}i\right)^{100}$ and $x = ky$, then k is

- (a) $-\frac{1}{3}$ (b) $\sqrt{3}$
(c) $-\sqrt{3}$ (d) $-\frac{1}{\sqrt{3}}$

123. If $z^2 - z + 1 = 0$, then $z^n - z^{-n}$, where n is a multiple of 3, is

- (a) $2(-1)^n$ (b) 0
(c) $(-1)^{n+1}$ (d) None of the above

124. If α is non-real and $\alpha = \sqrt[3]{1}$, then the value of $2^{1+\alpha+\alpha^2+\alpha^{-2}+\alpha^{-1}}$ is equal to

- (a) 4 (b) $1/2$
(c) 2 (d) None of these

125. If ω is a non-real cube root of unity, then the expression $(1 - \omega)(1 - \omega^2)(1 + \omega^4)(1 + \omega^8)$ is equal to

- (a) 0 (b) 3
(c) 1 (d) 2



Complex Number DPP

126. If ω is a non-real cube root of unity, then

$$\frac{1+2\omega+3\omega^2}{2+3\omega+\omega^2} + \frac{2+3\omega+\omega^2}{3+\omega+2\omega^2} \text{ is equal to}$$

- (a) -1 (b) 2ω
(c) 0 (d) -2ω

127. If $(\sqrt{3} + i)^n = (\sqrt{3} - i)^n, n \in N$, then least value of n is

- (a) 3 (b) 4
(c) 6 (d) None of these

128. If $x^3 - 1 = 0$ has the non-real complex roots α, β , then the value of $(1 + 2\alpha + \beta)^3 - (3 + 3\alpha + 5\beta)^3$ is

- (a) -7 (b) 6
(c) -5 (d) 0

129. If $z^2 - z + 1 = 0$, then the value of

$$\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 + \left(z^3 + \frac{1}{z^3}\right)^2$$

$+ \dots + \left(z^{24} + \frac{1}{z^{24}}\right)$ is equal to

- (a) 24 (b) 32
(c) 48 (d) None of these

130. When the polynomial $5x^3 + Mx + N$ is divided by $x^2 + x + 1$, the remainder is 0. The value of $(M + N)$ is equal to

- (a) -3 (b) 5
(c) -5 (d) 15

131. If $z = -2 + 2\sqrt{3}i$, then $z^{2n} + 2^{2n}z^n + 2^{4n}$ may be equal to

- (a) 2^{2n}
(b) 0
(c) $3 \cdot 4^{2n}, n$ is multiple of 3
(d) None of the above

132. Statement I: If $x + \frac{1}{x} = 1$ and $p = x^{4000} + \frac{1}{x^{4000}}$ and q is the digit at unit place in the number $2^{2n} + 1, n \in N$ and $n > 1$, then the value of $p + q =$

8.
Statement II: ω, ω^2 are the roots of $x + \frac{1}{x} = -1, x^3 + \frac{1}{x^3} = 2$

Which of the above statement(s) is/are correct ?

- (a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

133. If $\omega (\neq 1)$ is cube root of unity satisfying

$$\frac{1}{a+\omega} + \frac{1}{b+\omega} + \frac{1}{c+\omega} = 2\omega^2 \text{ and } \frac{1}{a+\omega^2} + \frac{1}{b+\omega^2} + \frac{1}{c+\omega^2} = 2\omega,$$

then the value of $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}$ is equal to

- (a) 2 (b) -2
(c) $-1 + \omega^2$ (d) None of these

134. If α and β are the roots of the equation $x^2 - 2x + 4 = 0$, then the value of $\alpha^6 + \beta^6$ is

- (a) 64 (b) 128
(c) 256 (d) None of these

135. If z is any complex number such that $z + \frac{1}{z} = 1$, then the value of $z^{99} + \frac{1}{z^{99}}$ is

- (a) 1 (b) -1
(c) 2 (d) -2

136. The value of $(-1 + \sqrt{-3})^{62} + (-1 - \sqrt{-3})^{62}$ is

- (a) 2^{62} (b) 2^{64}
(c) -2^{62} (d) 0

137. If ω is a cube root of unity, then

$(3 + 5\omega + 3\omega^2)^2 (3 + 3\omega + 5\omega^2)^2$ is equal to

- (a) 4 (b) 0
(c) -4 (d) None of these

138. The value of $\left(\frac{1+i\sqrt{3}}{1-i\sqrt{3}}\right)^6 + \left(\frac{1-i\sqrt{3}}{1+i\sqrt{3}}\right)^6$ is

- (a) 2 (b) -2
(c) 1 (d) 0

139. If $z_1 = \sqrt{3} + i\sqrt{3}$ and $z_2 = \sqrt{3} + i$, then the complex number $\left(\frac{z_1}{z_2}\right)^{50}$ lies in the quadrant number

- (a) I (b) II
(c) III (d) IV

140. If $x = a + b, y = a\omega + b\omega^2, z = a\omega^2 + b\omega$, then xyz is equal to

- (a) $(a + b)^3$ (b) $a^3 + b^3$
(c) $a^3 - b^3$ (d) $(a + b)^3 + 3ab(a + b)$

141. If 1, ω and ω^2 are the cube roots of unity, then the value of $(1 + \omega)^3 - (1 + \omega^2)^3$ is

- (a) 2ω (b) 2
(c) -2 (d) 0

142. If 1, ω and ω^2 are the three cube roots of unity and α, β and γ are the roots of $p, p < 0$, then



Complex Number DPP

for any x, y and z , the expression $\frac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha}$ equals

- (a) 1 (b) ω
(c) ω^2 (d) None of the above

143. If α and β are the complex cube roots of unity, then $\alpha^3 + \beta^3 + \alpha^{-2}\beta^{-2}$ is equal to

- (a) 0 (b) 3
(c) -3 (d) None of these

144. If ω is a complex cube root of unity, then $(1 - \omega + \omega^2)(1 - \omega^2 + \omega^4)(1 - \omega^4 + \omega^8)(1 - \omega^8 + \omega^{16})$ is equal to

- (a) 12 (b) 14
(c) 16 (d) None of these

145. If $x = \omega - \omega^2 - 2$, then the value of $x^4 + 3x^3 + 2x^2 - 11x - 6$ is

- (a) 1 (b) -1
(c) 2 (d) None of these

146. If $1, \omega$ and ω^2 are the three cube roots of unity, then $(1 + \omega)(1 + \omega^2)(1 + \omega^4)(1 + \omega^8) \dots$ to $2n$ factors is equal to

- (a) 1 (b) -1
(c) 0 (d) None of these

147. The common roots of the equations $z^3 + 2z^2 + 2z + 1 = 0$ and $z^{1985} + z^{100} + 1 = 0$ are

- (a) -1, ω (b) -1, ω^2
(c) ω, ω^2 (d) None of the above

148. The values of $(16)^{1/4}$ are

- (a) $\pm 2, \pm 2i$ (b) $\pm 4, \pm 4i$
(c) $\pm 1, \pm i$ (d) None of the above

149. If p, q and r are positive integers and ω is an imaginary cube root of unity and $f(x) = x^{3p} + x^{3q+1} + x^{3r+2}$, then $f(\omega)$ is equal to

- (a) ω (b) $-\omega^2$
(c) 0 (d) None of the above

150. If $\omega (\neq 1)$ is a cube root of unity and $(1 + \omega)^7 = A + B\omega$. Then, (A, B) equals

- (a) (1, 1) (b) (1, 0)
(c) (-1, 1) (d) (0, 1)

151. If α and β are the roots of the equation $x^2 - x + 1 = 0$, then $\alpha^{2009} + \beta^{2009}$ is equal to

- (a) -2 (b) -1
(c) 1 (d) 2

152. If $z^2 + z + 1 = 0$, where z is complex number, then the value of $\left(z + \frac{1}{z}\right)^2 + \left(z^2 + \frac{1}{z^2}\right)^2 +$

$\left(z^3 + \frac{1}{z^3}\right)^2 + \dots + \left(z^6 + \frac{1}{z^6}\right)^2$ is

- (a) 54 (b) 6
(c) 12 (d) 18

153. If the cube roots of unity are $1, \omega, \omega^2$, then the roots of the equation $(x - 1)^3 + 8 = 0$, are

- (a) -1, $1 + 2\omega, 1 + 2\omega^2$
(b) -1, $1 - 2\omega, 1 - 2\omega^2$
(c) -1, -1, -1
(d) -1, $-1 + 2\omega, -1 - 2\omega^2$

154. If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ equals

- (a) 128ω (b) -128ω
(c) $128\omega^2$ (d) $-128\omega^2$

155. If ω, ω^2 are the cube roots of unity, then roots of equation $(x - 1)^3 + 25 = 0$ are

- (a) -5, $-5\omega, -5\omega^2$
(b) -4, $1 - 5\omega, 1 - 5\omega^2$
(c) 6, $1 - 5\omega, 1 + 5\omega^2$
(d) None of the above

156. The value of $(1 + \sqrt{3}i)^4 + (1 - \sqrt{3}i)^4$ is

- (a) -16 (b) 16
(c) 14 (d) -14

157. If $\left(\frac{3}{2} + i\frac{\sqrt{3}}{2}\right)^{50} = 3^{25}(x + iy)$, where x and y are real, then the ordered pair (x, y) is

- (a) (-3, 0) (b) (0, 3)
(c) (0, -3) (d) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

158. If a and b are real numbers such that $(2 + \alpha)^4 = a + b\alpha$, where $\alpha = \frac{-1+i\sqrt{3}}{2}$, then $a + b$ is equal to

- (a) 24 (b) 33
(c) 9 (d) 57

159. The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is

- (a) -2^{15} (b) $2^{15}i$
(c) $-2^{15}i$ (d) 6^5



160. If $i = \sqrt{-1}$, then $4 + 5\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{334} + 3\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)^{365}$ is equal to

- (a) $1 - i\sqrt{3}$ (b) $-1 + i\sqrt{3}$
(c) $i\sqrt{3}$ (d) $-i\sqrt{3}$

161. $\sqrt{-1 - \sqrt{1 - \sqrt{1 - \dots \infty}}}$ is equal to

- (a) 1 (b) -1
(c) ω^2 (d) $-\omega$

“Locus of Complex Number”

162. The set of values of k for which the equation $z\bar{z} + (-3 + 4i)\bar{z} - (3 + 4i)z + k = 0$ represents a circle is

- (a) $(-\infty, 25)$ (b) $(25, \infty)$
(c) $(5, \infty)$ (d) $(-\infty, 5)$

163. The complex numbers $z = x + iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right| = 1$, lie on

- (a) the X-axis
(b) the straight line $y = 5$
(c) a circle passing through the origin
(d) None of the above

164. If the equation $|z - z_1|^2 + |z - z_2|^2 = k$ represents the equation of a circle, where $z_1 = 2 + 3i, z_2 = 4 + 3i$ are the extremities of a diameter, then the value of k is

- (a) $\frac{1}{4}$ (b) 4
(c) 2 (d) None of these

165. The locus of z given by $\left|\frac{z-1}{z-i}\right| = 1$ is

- (a) a circle (b) an ellipse
(c) a straight line (d) a parabola

166. If $z = x + iy$ and

$|z - 2 + i| = |z - 3 - i|$, then locus of z is

- (a) $2x + 4y - 5 = 0$ (b) $2x - 4y - 5 = 0$
(c) $x + 2y = 0$ (d) $x - 2y + 5 = 0$

167. The equation

$$|z - i| + |z + i| = k, k > 0$$

can represent an ellipse, if k^2 is

- (a) < 1 (b) < 2
(c) > 4 (d) None of these

168. The equation $|z + i| - |z - i| = k$ represents a hyperbola if

- (a) $-2 < k < 2$ (b) $k > 2$
(c) $0 < k < 2$ (d) None of these

169. $\left|\frac{z-1}{z+1}\right| = 1$, represents

- (a) a circle (b) an ellipse
(c) a straight line (d) None of these

170. $|z - 4| < |z - 2|$, represents the region given by

- (a) $\operatorname{Re}(z) > 0$ (b) $\operatorname{Re}(z) < 0$
(c) $\operatorname{Re}(z) > 2$ (d) None of these

171. If $\operatorname{Re}\left(\frac{z+4}{2z-i}\right) = \frac{1}{2}$, then z is represented by a point lying on

- (a) a circle (b) an ellipse
(c) a straight line (d) None of these

172. If z is a complex number, then $z^2 + \bar{z}^2 = 2$ represents

- (a) a circle (b) a straight line
(c) a hyperbola (d) an ellipse

173. The equation $|z + 1 - i| = |z + i - 1|$ represents

- (a) a straight line (b) a circle
(c) a parabola (d) a hyperbola

174. If z_1, z_2 and z_3 are three complex numbers in AP, then they lie on

- (a) a circle (b) a straight line
(c) a parabola (d) an ellipse

175. If $\operatorname{Re}\left(\frac{z+2i}{z+4}\right) = 0$, then z lies on a circle with centre

- (a) $(-2, -1)$ (b) $(-2, 1)$
(c) $(2, -1)$ (d) $(2, 1)$

176. If $\operatorname{Im}\left(\frac{z+2i}{z+2}\right) = 0$, then z lies on the curve

- (a) $x^2 + y^2 + 2x + 2y = 0$
(b) $x^2 + y^2 - 2x = 0$
(c) $x + y + 2 = 0$
(d) None of the above



Complex Number DPP

177. If $z = x + iy$ and a is a real number such that $|z - ai| = |z + ai|$, then locus of z is

- (a) X -axis (b) Y -axis
(c) $x = y$ (d) $x^2 + y^2 = 1$

178. If P represents $z = x + iy$ in the argand plane and $|z - 1|^2 + |z + 1|^2 = 4$, then the locus of P is

- (a) $x^2 + y^2 = 2$ (b) $x^2 + y^2 = 1$
(c) $x^2 + y^2 = 4$ (d) $x + y = 2$

179. The locus of z satisfying $\text{Im}(z^2) = 4$ is

- (a) a circle
(b) a rectangular hyperbola
(c) a pair of straight lines
(d) None of the above

180. The curve represented by $\text{Re}(z^2) = 4$ is

- (a) a parabola (b) an ellipse
(c) a circle (d) a rectangular hyperbola

181. If $w = \frac{z}{z-3}$ and $|w| = 1$, then z lies on

- (a) a parabola (b) a circle
(c) a straight line (d) an ellipse

182. The complex number $z = x + iy$, which satisfies the equation $\left|\frac{z-3i}{z+3i}\right| = 1$, lie on

- (a) the X -axis
(b) the straight line $y = 3$
(c) a circle passing through the origin
(d) None of the above

183. If $\left|\frac{z-2}{z-3}\right| = 2$ represents a circle, then centre and radius are

- (a) $\left(\frac{10}{3}, 0\right), \frac{2}{3}$ (b) $\left(-\frac{10}{3}, 0\right), \frac{2}{3}$
(c) $\left(\frac{10}{3}, 1\right), \frac{2}{3}$ (d) None of these

“Minimum and Maximum Value”

184. For a complex number z , the minimum value of $|z| + |z - 2|$ is

- (a) 1 (b) 2
(c) 3 (d) None of these

185. If $|z_1 - 1| < 1$, $|z_2 - 2| < 2$ and $|z_3 - 3| < 3$, then $|z_1 + z_2 + z_3|$

- (a) is less than 6 (b) is more than 3
(c) is less than 12 (d) lies between 6 and 12

186. The maximum and minimum values of $|z + 1|$, when $|z + 3| \leq 3$ are

- (a) (5,0) (b) (6,0)
(c) (7,1) (d) (5,1)

187. The minimum value of $|1 + z| + |1 - z|$, where z is a complex number, is

- (a) 2 (b) $\frac{3}{2}$
(c) 1 (d) 0

188. If $\left|z + \frac{4}{z}\right| = a$, then find the greatest and least values of $|z|$.

189. If z is any complex number such that $|z + 4| \leq 3$, then the least value and greatest value of $|z + 1|$ are

- (a) 1,6 (b) 0,6
(c) 2,8 (d) None of these

190. For any complex number z , the minimum value of $|z| + |z - 1|$ is

- (a) 1 (b) 0
(c) $\frac{1}{2}$ (d) $\frac{3}{2}$

191. The maximum value of $|z|$ when z satisfies the condition $\left|z + \frac{2}{z}\right| = 2$ is

- (a) $\sqrt{3} - 1$ (b) $\sqrt{3} + 1$
(c) $\sqrt{3}$ (d) $\sqrt{2} + \sqrt{3}$

192. If $|z_1| = 15$ and $|z_2 - 3 - 4i| = 5$, then which of the following is correct ?

- (1) $|z_1 - z_2|_{\min} = 5$ (2) $|z_1 - z_2|_{\min} = 10$
(3) $|z_1 - z_2|_{\max} = 20$ (4) $|z_1 - z_2|_{\max} = 25$

193. If $|z - (1/z)| = 1$, then

which of the following statement is correct ?

- (1) $|z|_{\max} = \frac{1+\sqrt{5}}{2}$ (2) $|z|_{\min} = \frac{\sqrt{5}-1}{2}$
(3) $|z|_{\max} = \frac{\sqrt{5}-2}{2}$ (4) $|z|_{\min} = \frac{\sqrt{5}-1}{\sqrt{2}}$

194. If z is a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{z}\right|$

- (a) is equal to $\frac{5}{2}$
(b) lies in the interval (1,2)
(c) is strictly greater than $\frac{5}{2}$
(d) is strictly greater than $\frac{3}{2}$ but less than $\frac{5}{2}$



195. If $\left|z - \frac{4}{z}\right| = 2$, then the maximum value of $|z|$ is equal to

- (a) $\sqrt{3} + 1$ (b) $\sqrt{5} + 1$
(c) 2 (d) $2 + \sqrt{2}$

196. The complex number z satisfies the condition

$\left|z - \frac{25}{z}\right| = 24$. The maximum distance from the origin of coordinates to the point z is

- (a) 25 (b) 30
(c) 32 (d) None of these

Two Important Points :

(i) The area of triangle whose vertices are z, iz and $z + iz$ is $\frac{1}{2}|z|^2$.

(ii) The area of triangle whose vertices are $z, \omega z$ and $z + \omega z$ is $\frac{\sqrt{3}}{4}|z|^2$.

Example :

197. If the area of the triangle on the complex plane formed by the points z, iz and $z + iz$ is 50 sq units, then $|z|$ is

- (a) 5 (b) 10
(c) 15 (d) None of these

198. If the area of the triangle on the complex plane formed by complex numbers $z, \omega z$ and $z + \omega z$ is $4\sqrt{3}$ sq units, then $|z|$ is

- (a) 4 (b) 2
(c) 6 (d) 3

